PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 9 · Storytelling and visualization

TUESDAY OCT 20 AND THURSDAY OCT 22, 2026

SETUP

UPDATE BOTH FOLDERS BEFORE WE START

1 Open Claude Code in your repo folder, spax402-your-username.

Your personal repository sits next to spax402-course-materials, the shared folder Professor Davis updates weekly.

2 Run one prompt.

The /update skill brings this week's files into your personal repo without saving over your work, and the course materials folder updates by asking Claude to pull the latest version.

> Run /update to refresh my repo, then pull the latest course materials in the folder next to mine, spax402-course-materials. ↵

3 Confirm that today's deck is in spax402-course-materials/decks/.

Thursday's files arrive through /update before Thursday's class.

01

EXPLAINING, NOT EXPLORING

EXPLORATORY VS EXPLANATORY ANALYSIS

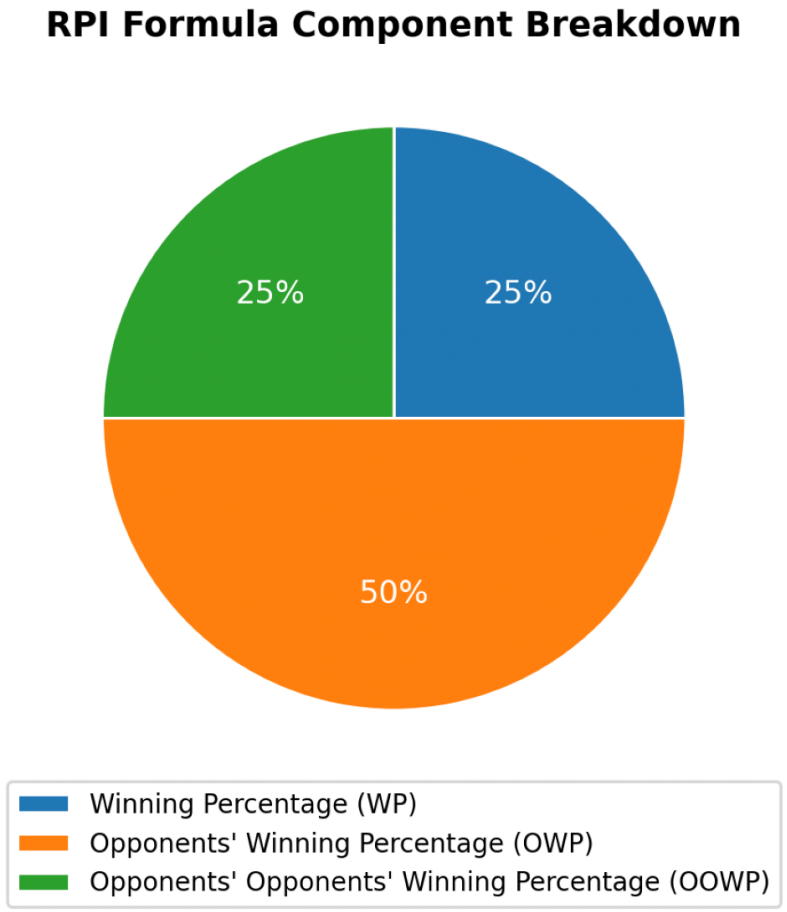
EXPLORATORY

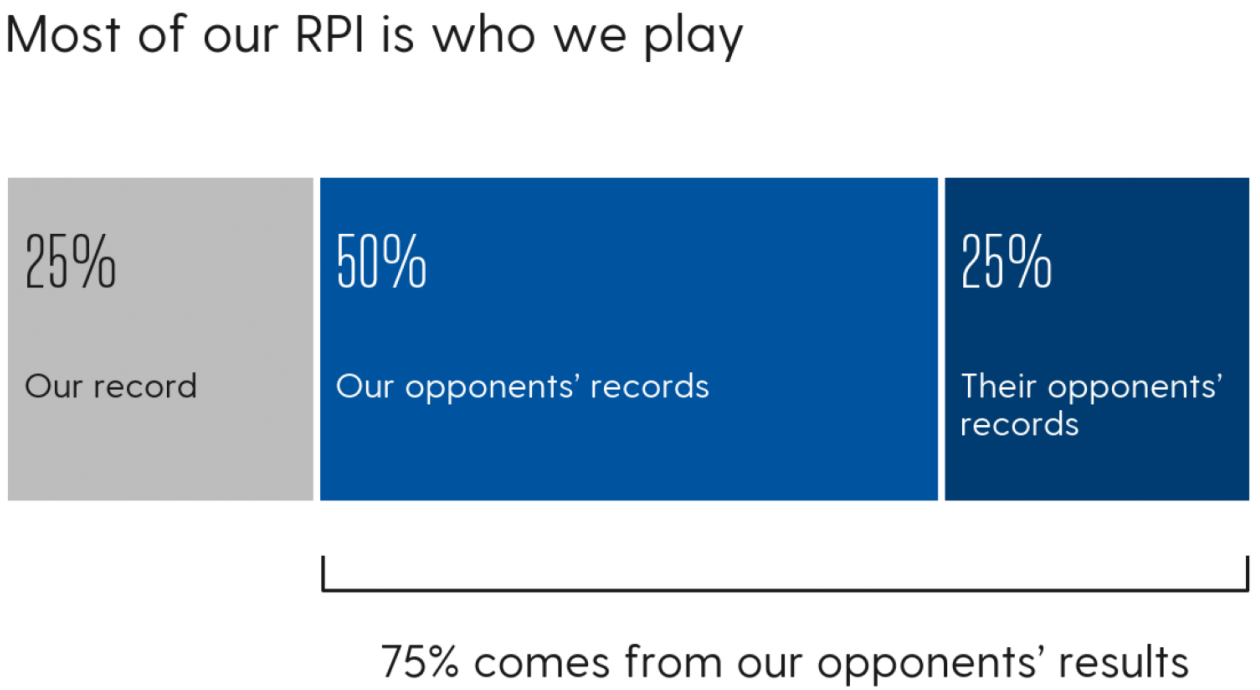
For the analyst. Comprehensive. To discover.

EXPLANATORY

For the audience. Distilled. To persuade.

WHICH ONE EXPLAINS RPI BETTER TO OUR HEAD COACH?





The NCAA RPI formula. Public.

“I WOULD’VE WRITTEN A SHORTER LETTER, BUT I DID NOT HAVE THE TIME.”

RECOMMENDED READING

Nussbaumer Knaflic, C. (2015). Storytelling with Data: A Data Visualization Guide for Business Professionals. Hoboken, NJ: John Wiley & Sons.

02

KNOW THE AUDIENCE

UNDERSTAND THE CONTEXT

1. Who is your audience?
2. Do they trust you or do you have to build credibility?
3. Do they want to hear the process or just the bottom line?
4. Do they value nuance or simplicity?

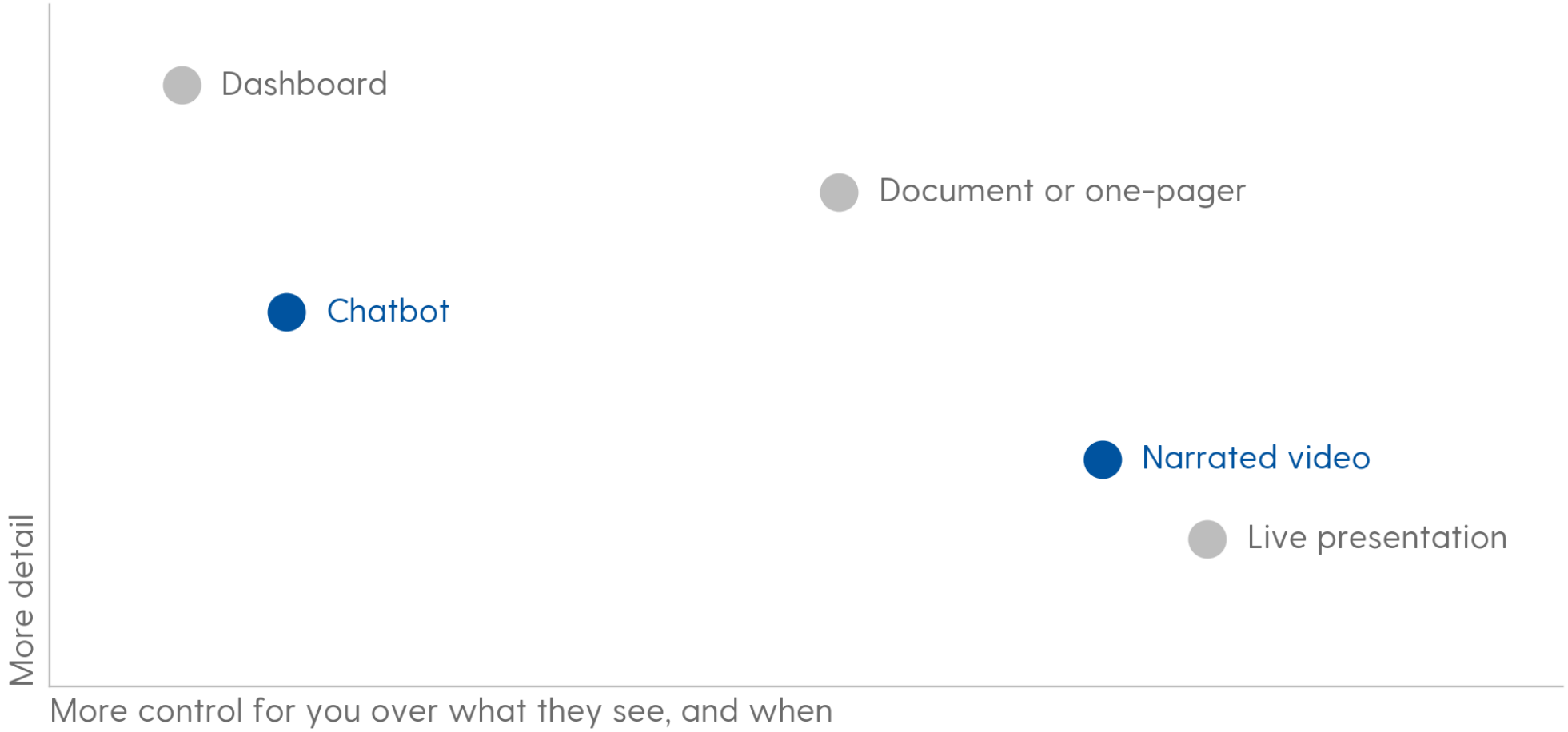
Source: Nussbaumer Knaflic, Storytelling with Data (2015).

WHAT DO YOU NEED YOUR AUDIENCE TO KNOW OR DO?

Some audiences want a recommended action. Others know what to do better than we do and want the insight. Then we're the tech reading the MRI, not the doctor writing the treatment plan.

? POST-SEASON RECAP: AN ATHLETIC DIRECTOR, A HEAD COACH AND A DONOR PROSPECT. WHAT DOES EACH NEED? HOW WOULD YOU DELIVER THE INFO?

CONTROL AND DETAIL



CHOOSING THE MEDIUM

MEDIUM	WHO SETS THE PACE	DETAIL	BEST WHEN
Live presentation	You	Lowest	You need a decision in the room
Narrated video	You, without being there	Low	A busy audience, on their own time
Document or one-pager	The reader	High	They'll read it later and want the detail
Dashboard	The reader, exploring	Highest	They have their own questions and time to ask them
Chatbot	The reader, by asking	On demand	They have questions you can’t predict and won’t open a dashboard

Live presentation and document follow Knaflic’s control and detail trade-off.

? KNAFLIC SAYS TO GIVE YOUR AUDIENCE A RECOMMENDED ACTION. IN MY SKILL DOCUMENT I FORBID OUR FOOTBALL VIDEOS FROM DOING THAT. WHICH IS THE RIGHT APPROACH?

03

FOCUS ATTENTION

COUNT THE NUMBER OF 3'S.

7 9 5 4 2 2 0 5 8 7 9 1 5 8 9 0 6 2 7 6 2 2 3 0 1 2

8 9 1 6 1 4 3 3 6 1 4 3 6 4 5 0 3 0 6 0 6 7 2 0 3 6

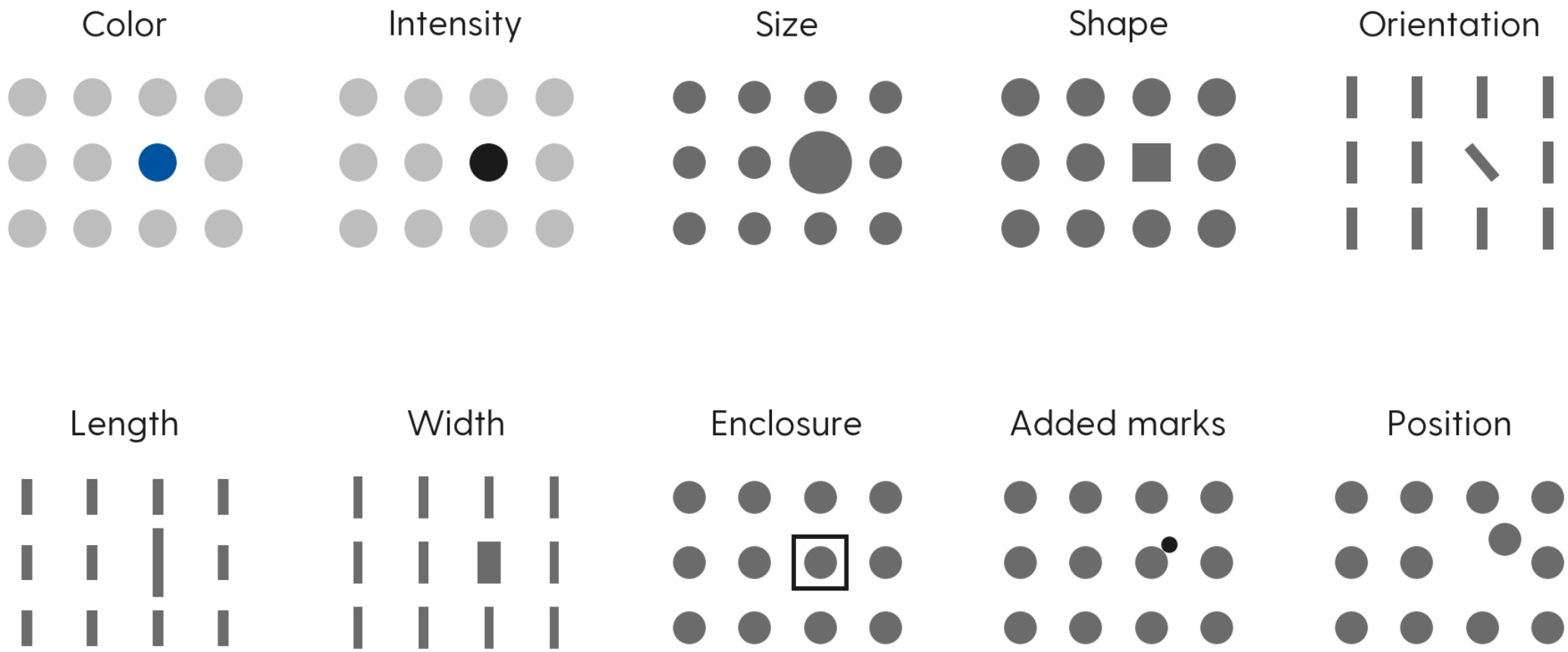
1 9 0 1 9 3 3 5 0 1 2 8 0 8 1 9 7 8 3 6 1 6 3 1 9 2

2 9 0 0 4 8 9 2 4 9 0 1 6 3 2 9 8 0 5 8 9 9 1 5 1 9

5 7 3 6 3 9 0 6 9 5 0 6 1 3 3 7 4 5 1 4 4 1 8 0 0 6

8 6 7 5 3 9 1 6 3 1 6 8 5 8 5 2 4 2 4 0 0 2 4 2 1 9

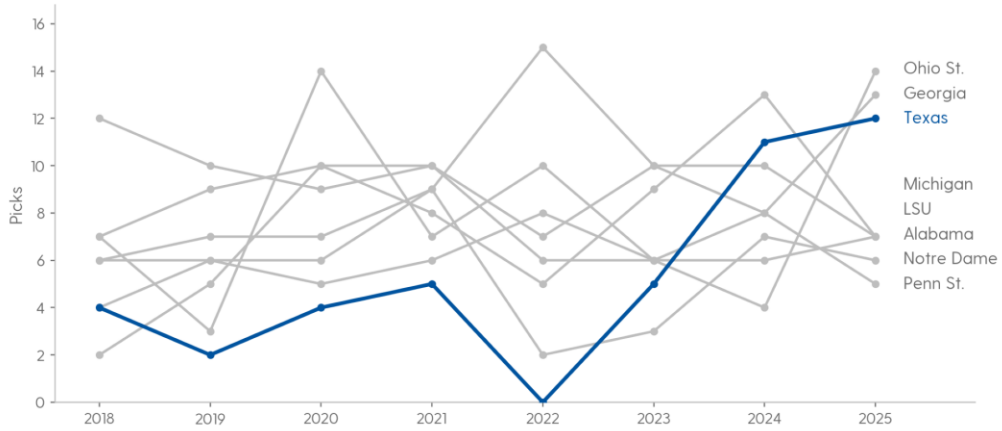
PRE-ATTENTIVE ATTRIBUTES

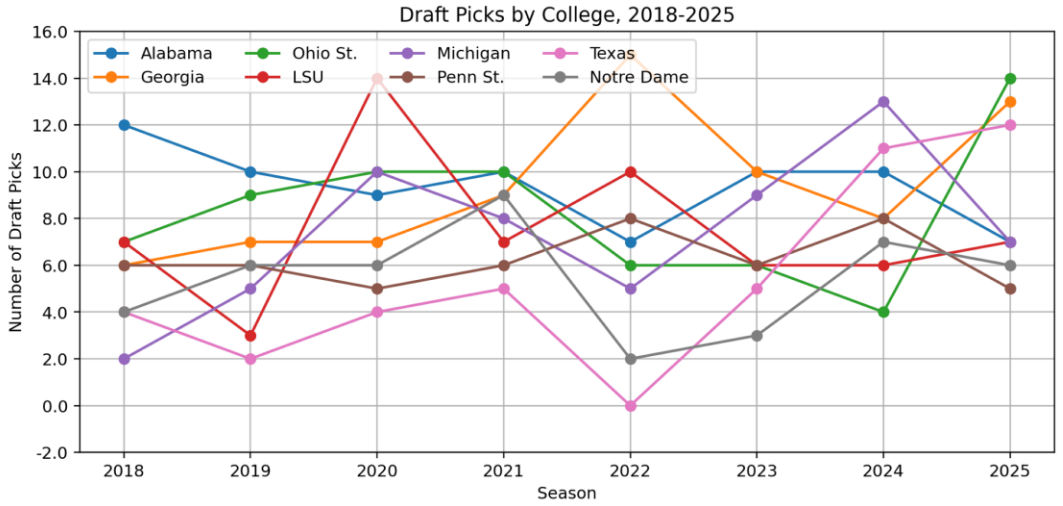


Source: Nussbaumer Knaflic, Storytelling with Data (2015).

WHICH WOULD YOU SEND A DIRECTOR OF COLLEGE SCOUTING?

Texas went from no draft picks in 2022 to 23 across the last two drafts





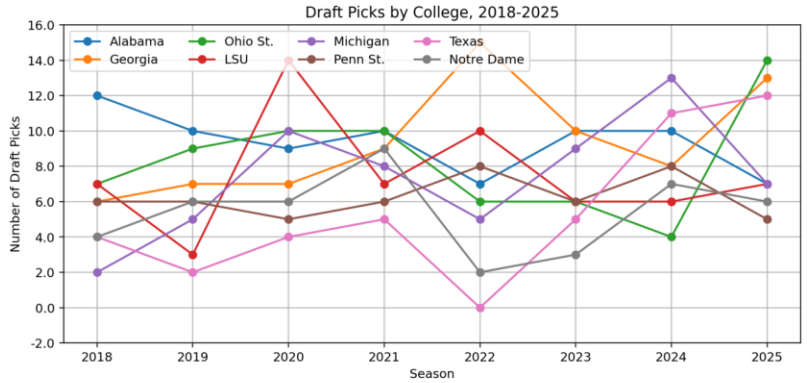
Source: nflverse draft_picks release file, 2018 to 2025 drafts.

MAXIMIZE THE SHARE OF INK THAT SHOWS DATA.

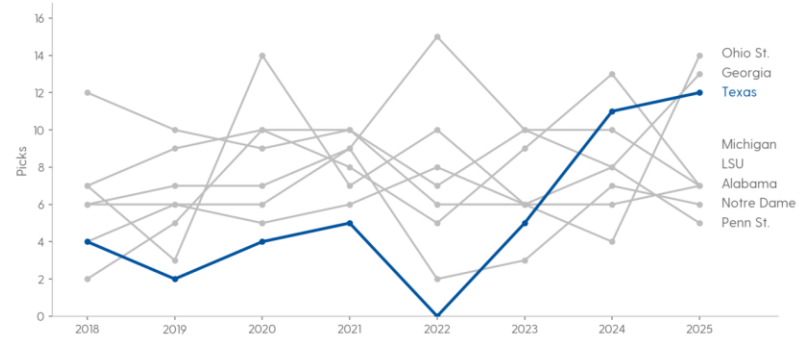
Data-ink ratio: the ink that carries values, divided by all the ink on the page.

Borders, gridlines, shading, legends and 3-D effects are candidates to cut.

CLUTTER IS YOUR ENEMY



Texas went from no draft picks in 2022 to 23 across the last two drafts



Decimals

Label directly

Remove markers?

Remove gridlines?

Minimum needed axis labels

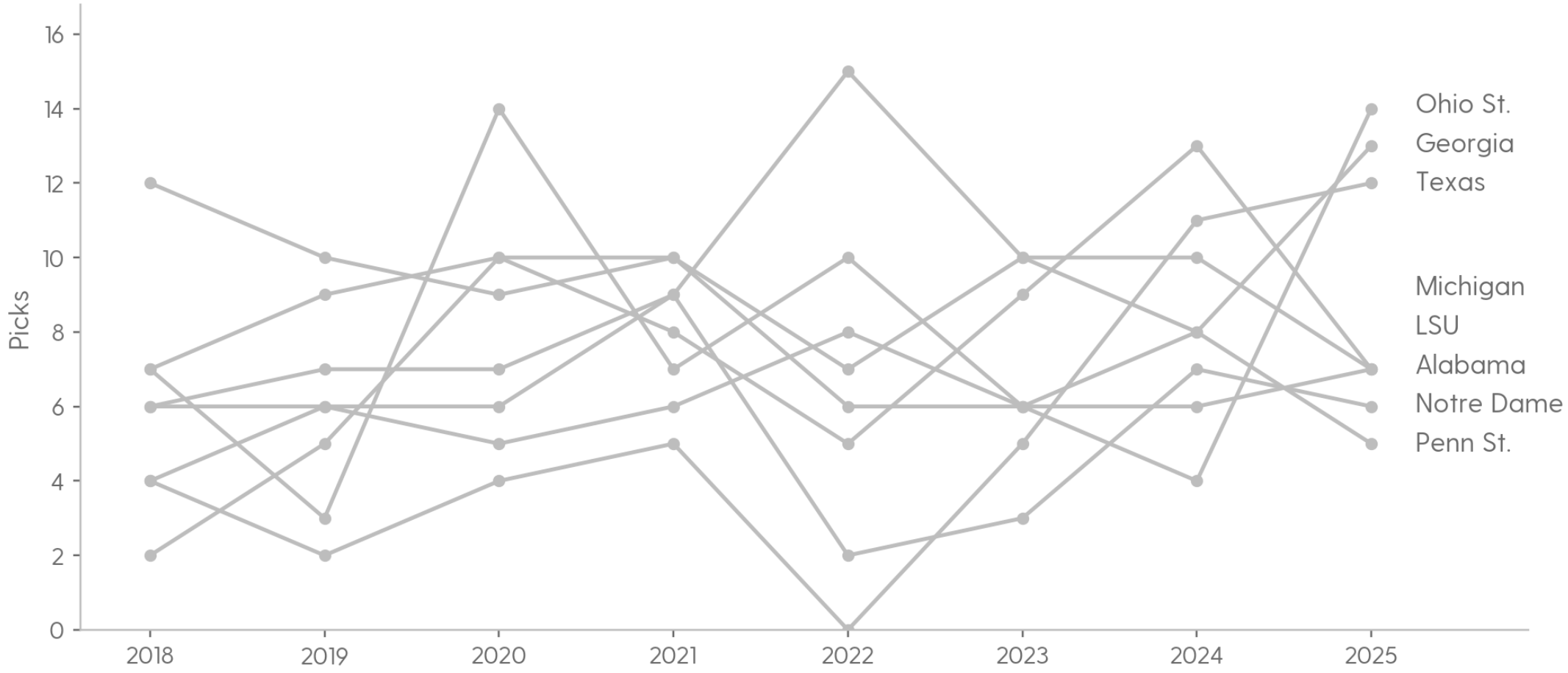
Abbreviate month names

Eliminate ticks

Remove chart border

Source: Nussbaumer Knaflic, Storytelling with Data (2015).

DRAFT PICKS BY PROGRAM, 2018 TO 2025



Source: nflverse draft_picks release file, 2018 to 2025 drafts.

04

CHOOSE THE FORM

MATCH THE CHART TO THE COMPARISON

YOU ARE COMPARING	USE
One or two numbers	A sentence with the number in it
Categories	Bars, horizontal when the names are long
Change over time	A line
Exact values someone will look up	A table
Parts of a whole, two or three parts	A stacked bar, with caution
Increases and decreases from a start to an end	A waterfall

Source: Nussbaumer Knaflic, Storytelling with Data (2015).

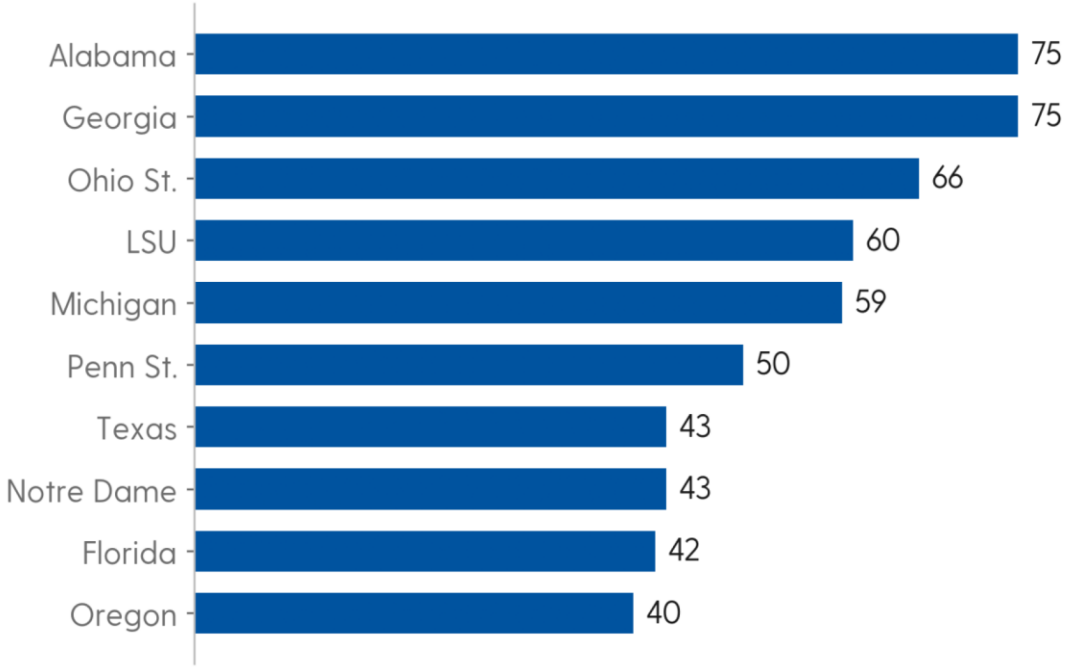
JUST BECAUSE YOU'RE COMMUNICATING A NUMBER, DOESN'T MEAN YOU NEED A GRAPH.

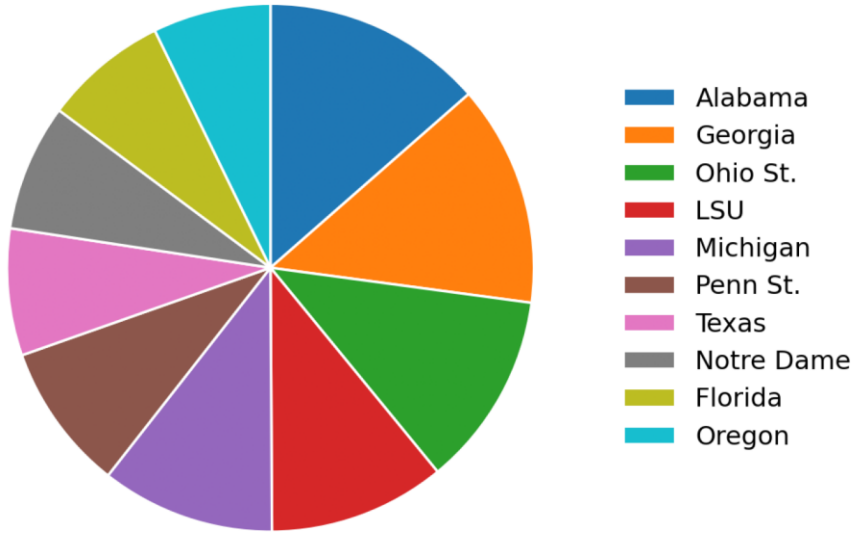
27%

of NFL draft picks from 2018 to 2025 came from 10 schools.

Source: nflverse draft_picks release file, 2018 to 2025 drafts.

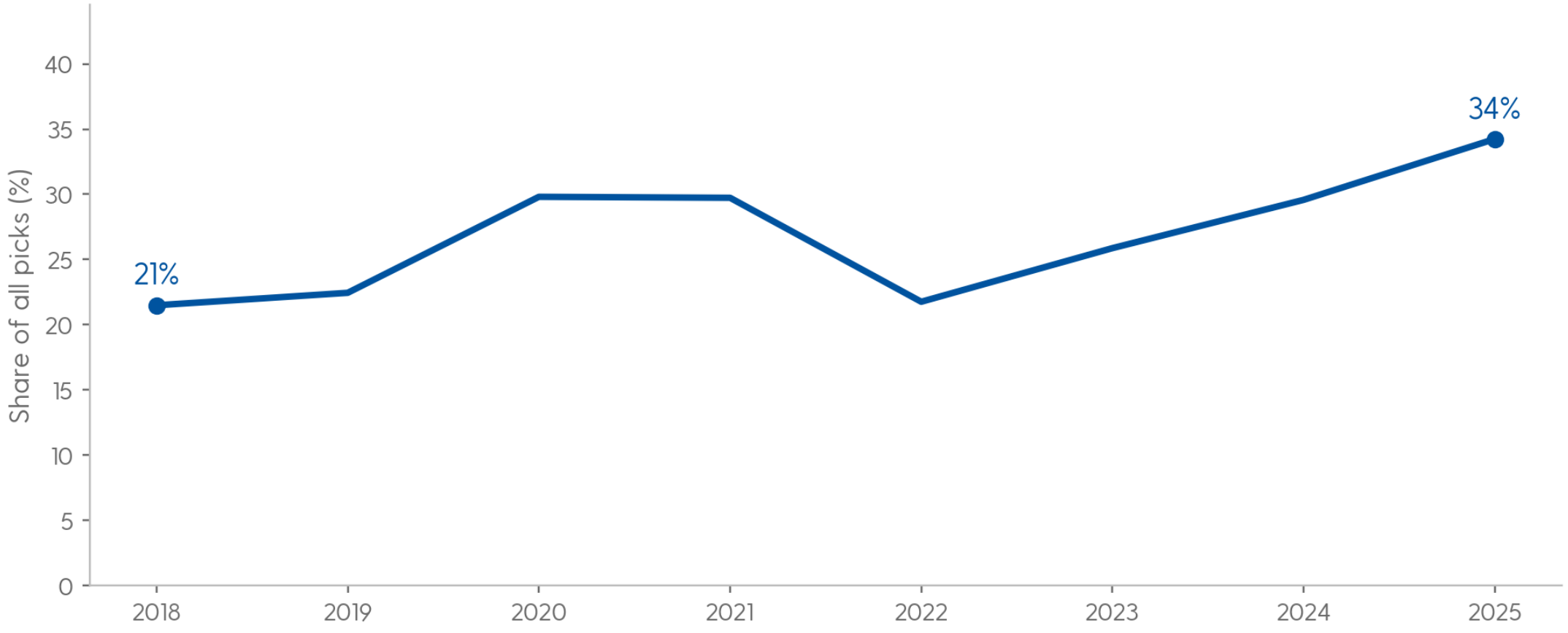
WHICH PROGRAMS PRODUCED THE MOST DRAFT PICKS?





Source: nflverse draft_picks release file, 2018 to 2025 drafts.

SHARE OF DRAFT PICKS FROM THE TOP 10 PROGRAMS, 2018 TO 2025



Source: nflverse draft_picks release file, 2018 to 2025 drafts.

THE DIRECTOR WANTS EACH PROGRAM'S PICKS BY YEAR. WHICH TABLE?

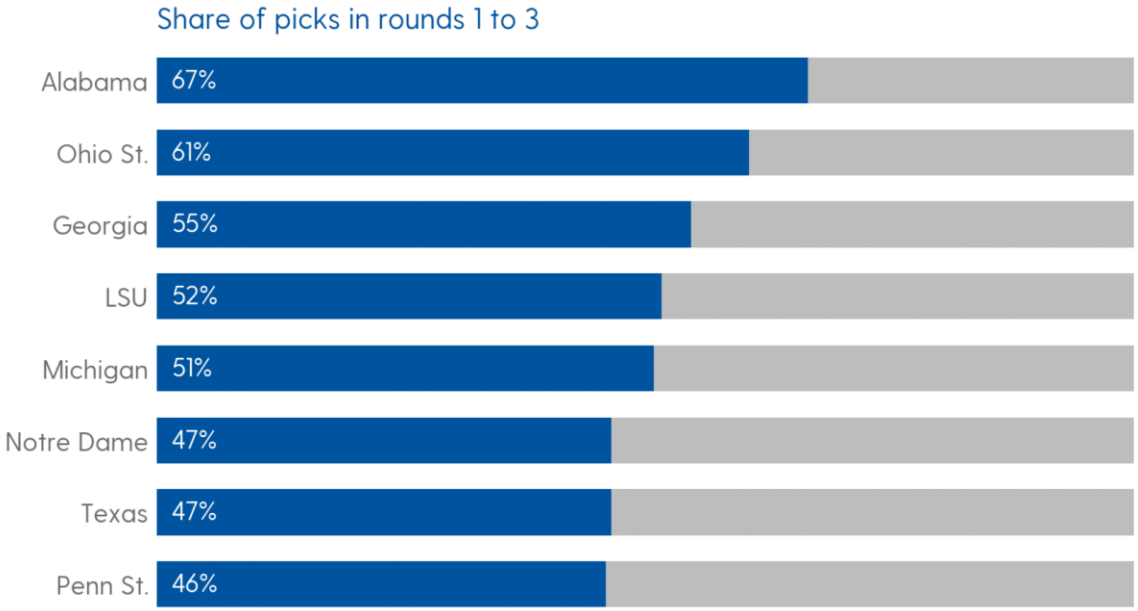
Draft Picks by College and Year (Heatmap)

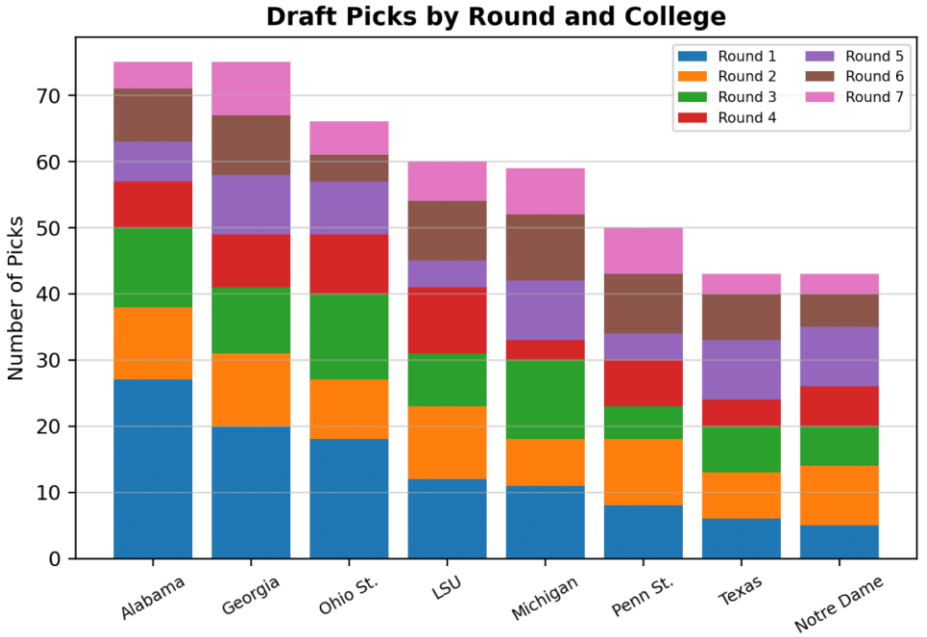
	2018	2019	2020	2021	2022	2023	2024	2025	Share
Alabama	12.00	10.00	9.00	10.00	7.00	10.00	10.00	7.00	15.9236%
Georgia	6.00	7.00	7.00	9.00	15.00	10.00	8.00	13.00	15.9236%
Ohio St.	7.00	9.00	10.00	10.00	6.00	6.00	4.00	14.00	14.0127%
LSU	7.00	3.00	14.00	7.00	10.00	6.00	6.00	7.00	12.7389%
Michigan	2.00	5.00	10.00	8.00	5.00	9.00	13.00	7.00	12.5265%
Penn St.	6.00	6.00	5.00	6.00	8.00	6.00	8.00	5.00	10.6157%
Texas	4.00	2.00	4.00	5.00	0.00	5.00	11.00	12.00	9.1295%
Notre Dame	4.00	6.00	6.00	9.00	2.00	3.00	7.00	6.00	9.1295%

	'18	'19	'20	'21	'22	'23	'24	'25	Total
Alabama	12	10	9	10	7	10	10	7	75
Georgia	6	7	7	9	15	10	8	13	75
Ohio St.	7	9	10	10	6	6	4	14	66
LSU	7	3	14	7	10	6	6	7	60
Michigan	2	5	10	8	5	9	13	7	59
Penn St.	6	6	5	6	8	6	8	5	50
Texas	**4**	**2**	**4**	**5**	**0**	**5**	**11**	**12**	**43**
Notre Dame	4	6	6	9	2	3	7	6	43

Source: nflverse draft_picks release file, 2018 to 2025 drafts.

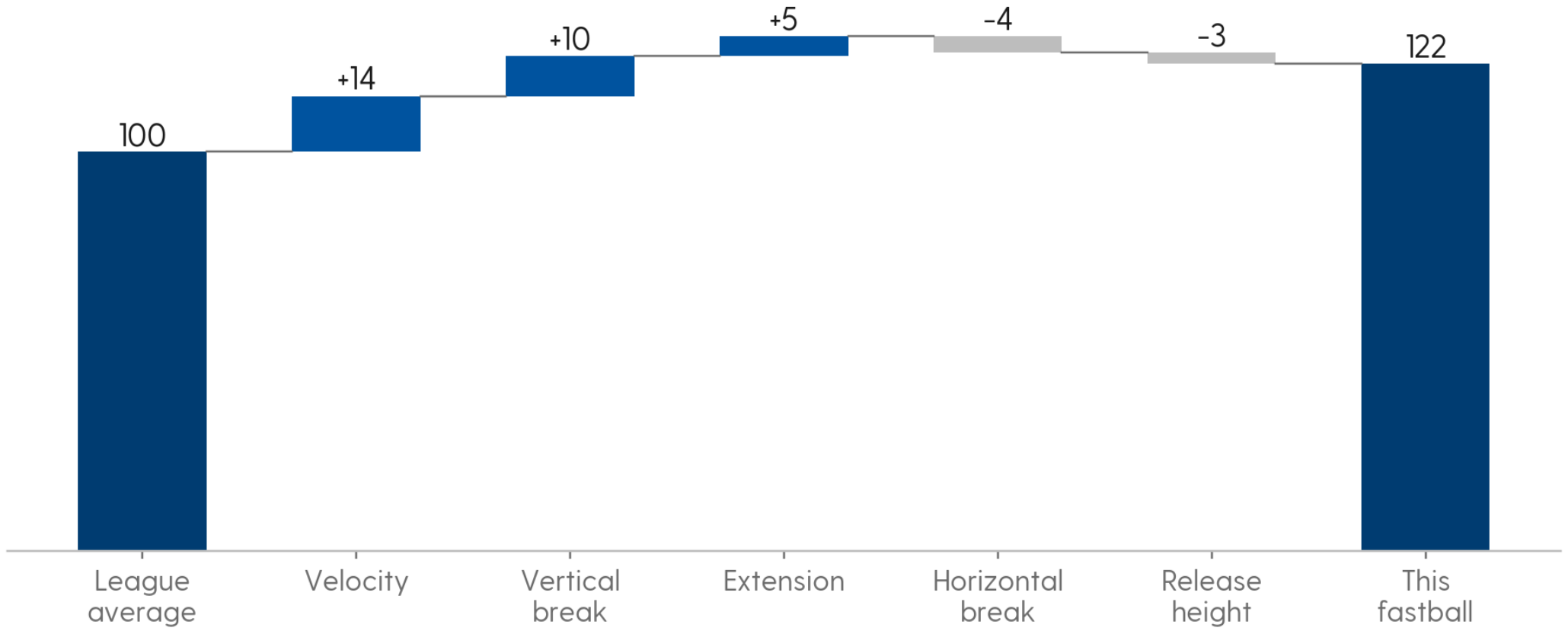
HOW EARLY DO EACH PROGRAM'S PLAYERS GO?





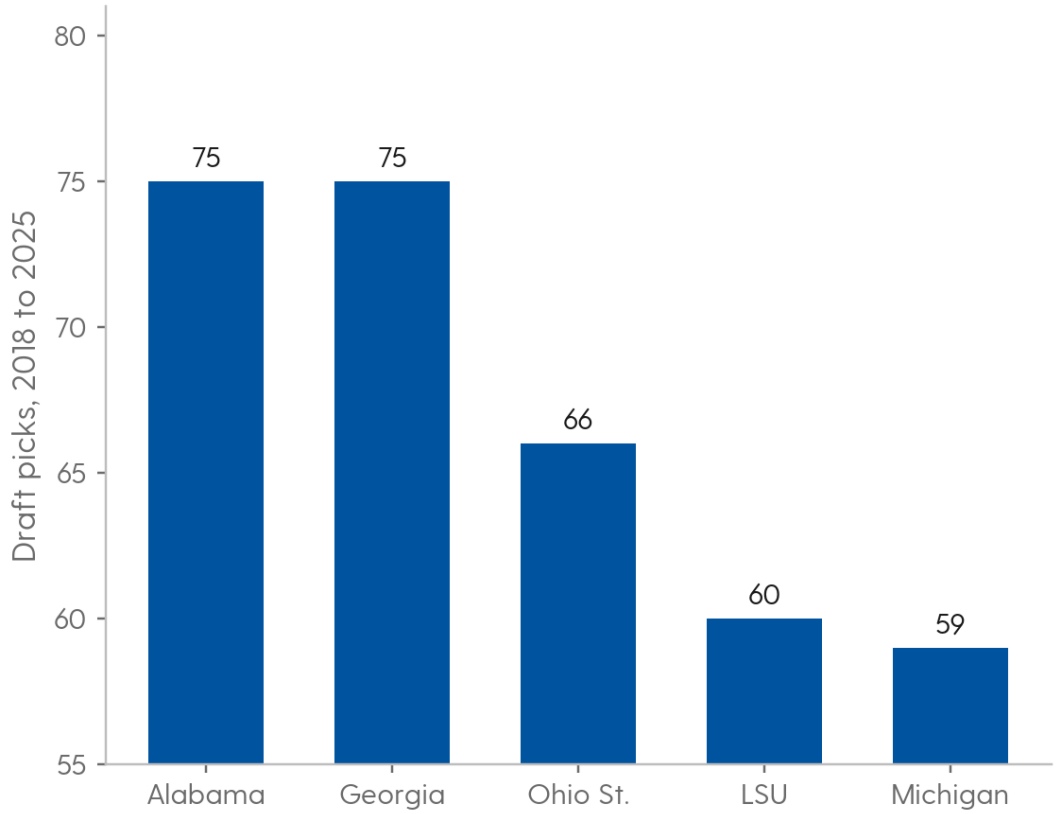
Source: nflverse draft_picks release file, 2018 to 2025 drafts.

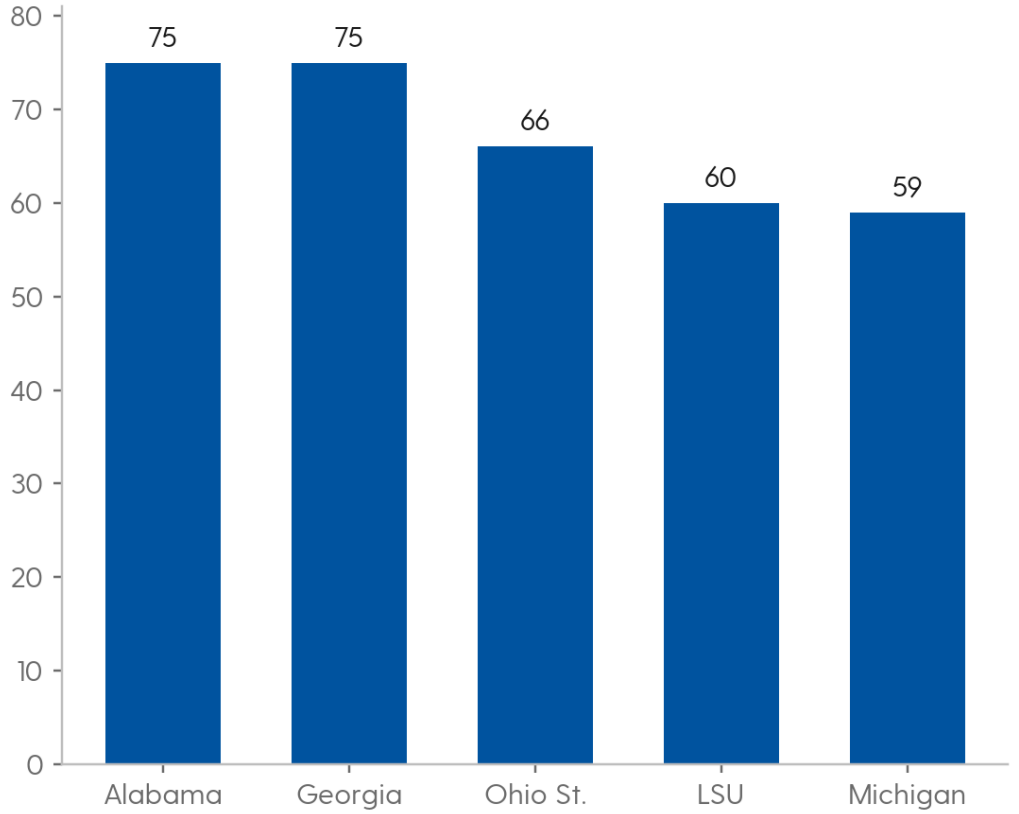
STUFF+ FOR ONE FASTBALL



Source: Illustrative values, not a real pitcher.

WHICH CHART BETTER COMPARES ALABAMA AND MICHIGAN?





Source: nflverse draft_picks release file, 2018 to 2025 drafts.

05

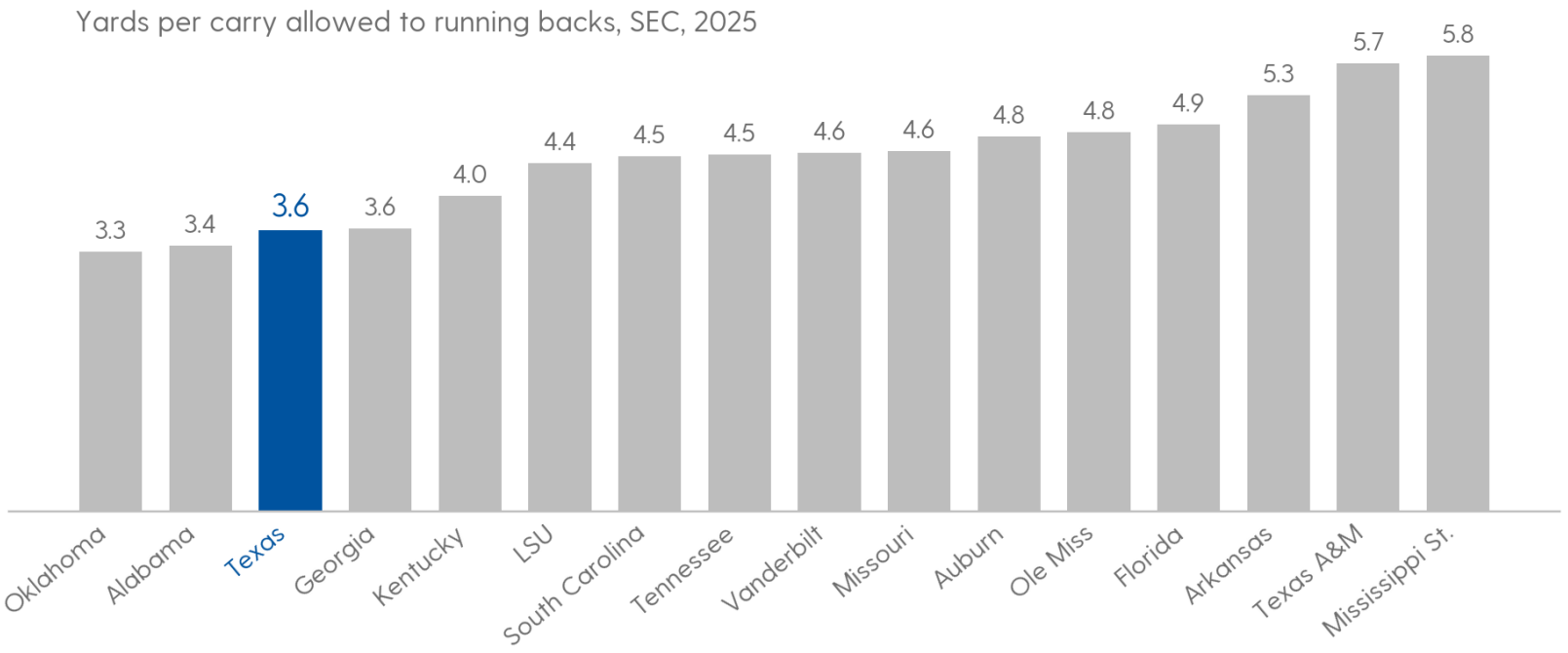
TELL A STORY

PEOPLE FORGET FACTS, BUT THEY REMEMBER STORIES.

TELL A STORY

1. Introduce a problem
2. Reinforcement & a strategy
3. A solution

A PROBLEM: TEXAS HELD RUNNING BACKS TO 3.6 YARDS PER CARRY, THIRD IN THE SEC.



The Middle: We need to find weaknesses in the opponent's personnel to put our players in a position to succeed.

Source: PFF, 2025 regular season and SEC title game.

30%

of #1's tackle attempts on quarterbacks were missed tackles.

30%

of #1's tackle attempts on quarterbacks were missed tackles.

Use repetition!

TEXAS HELD RUNNING BACKS TO 3.6 YARDS PER CARRY.

If we leave #1 unblocked and read him with zone read and quarterback keeps, he has to tackle our quarterback in space, and we will create explosive runs.

06

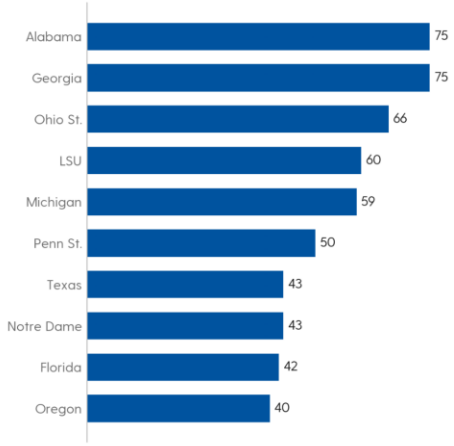
WHAT AI-MADE WORK LOOKS LIKE

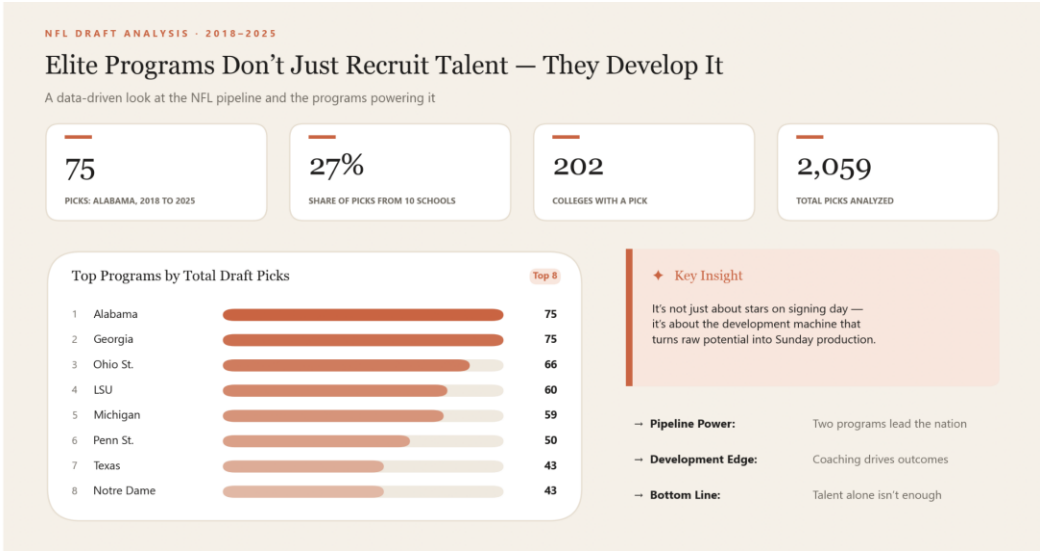
WHICH ONE DID A PERSON MAKE?

WHERE NFL DRAFT PICKS COME FROM

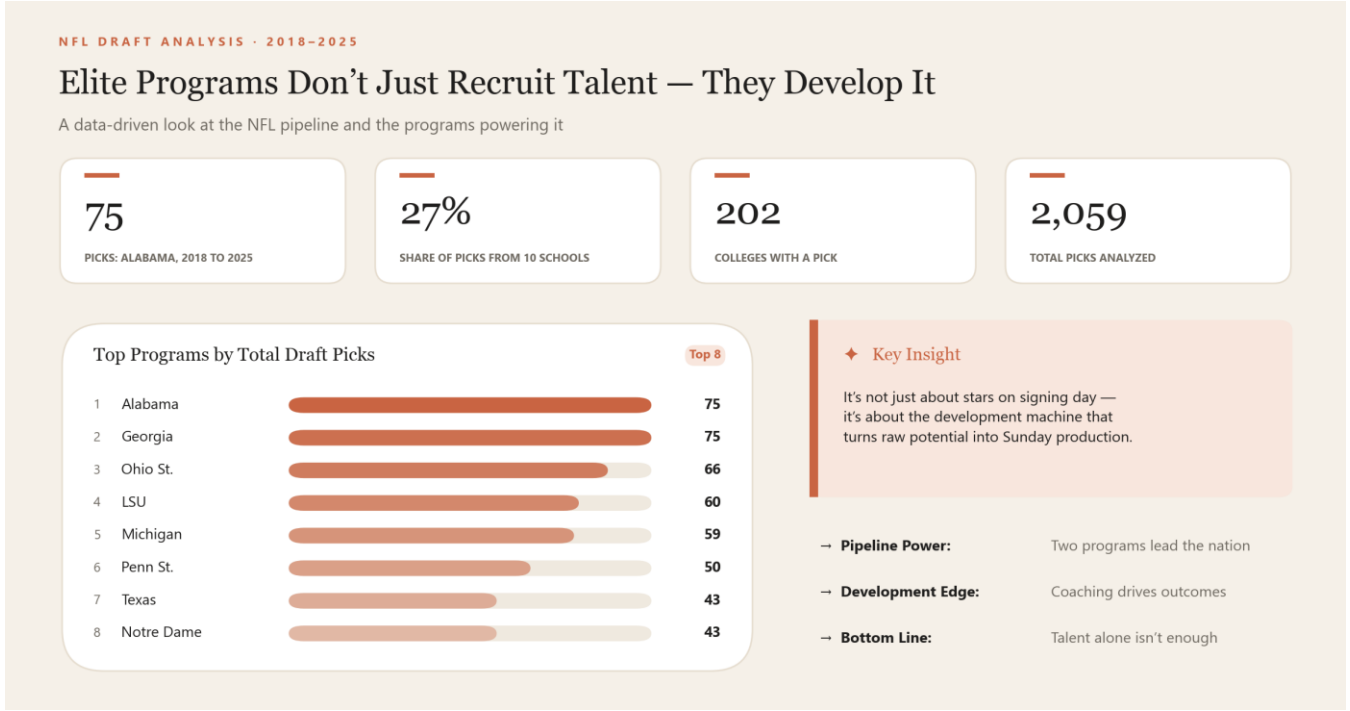
Ten programs produced 27% of the 2,059 players drafted from 2018 to 2025. Alabama and Georgia lead with 75 each.

Pick counts mix recruiting and development, so this doesn't show who develops players best.





A ONE-PAGER ON THE DRAFT



WHAT GIVES THIS AWAY AS AI-MADE?

Every number on it is real. Source: nflverse draft_picks release file, 2018 to 2025 drafts.

EVERY TELL IS A RULE YOU CAN WRITE DOWN.

Write it into your skill to tell Claude not to do it. You can also frame style positively or point it to ideal examples.

07

MODERN TOOLS

CLAUDE DESIGN, DASHBOARDS, MOTION AND SLIDES

1 **Design.**

Many layouts in minutes. Your job becomes choosing between them.

2 **Dashboards.**

Explorable views for a reader with their own questions.

3 **Motion.**

Animates your charts, text and images into short explainers.

4 **Slides.**

Allows you to collaborate in real-time with Claude.

MEN'S SOCCER SCHEDULING, NARRATED

DASHBOARD OR VIDEO: WHICH WOULD THE HEAD COACH ACTUALLY USE?

RULES FROM OUR GAME-WEEK VIDEOS

1 **Few words, big pictures.**

Simple, large takeaways come on screen as you say them.

2 **Real film plus simple drawings.**

Makes the data real for a coach.

3 **No statistics jargon.**

Statistics get us to the answer. They don’t make it into the presentation.

08

THIS WEEK'S WORK

WHAT A SKILL IS

1. **A file of instructions Claude loads when the task calls for it.**
   SKILL.md, in its own folder.

2. **The description says when to use it.**
   “Use when making a chart, one-pager or deck for a coach or front office.”

3. **The body says how.**
   The rules, the AI tells to avoid, and one good example.

4. **It applies every time.**
   A rule in a chat is gone next session.

RULES FILES WE ALREADY USE

1 **UD Athletics data visualization standard.**

"Direct labels always beat legends." Blue for the data, gray for context, one yellow highlight.

2 **The rules these decks are built from.**

Every one of them was an instinct that got corrected.

3 **The rules for anything a coach reads.**

Written after the men's soccer brief.

THURSDAY

WRITE YOUR SKILL AND TEST IT

1 Write it with Claude.

> Interview me one question at a time about how I want charts and one-pagers made for a coach or front office: audience, takeaway, chart form, color, clutter, and AI tells to avoid. Then save it as a skill called viz-story.

2 Run it on your Week 8 leaderboard.

> Use my viz-story skill to rebuild my Week 8 leaderboard chart for an NFL general manager or a college recruiting director. Save the old and new versions side by side.

3 Check it against your own rules.

> Review the new chart against every rule in my viz-story skill and make sure it complies. If needed, fix the skill along with the chart.

CASE STUDY

ONE STORY, TWO MEDIUMS

STEPS

1. Pick your audience: an NFL team's general manager or a college team's recruiting director. Build them a one-pager or a three-slide deck from your Week 8 leaderboard, using your skill.
2. Write a 60 to 90 second narration script that tells the same story.
3. Turn it into a video with your recorded voice narrating it.
4. Update skills with anything you had to fix by hand.

BRIEF QUESTIONS

1. What is the one thing you want the audience to take away? What did you intentionally leave out?
2. What did your skill fix in your first draft? Did it miss anything that you had to fix yourself?
3. Would your audience rather get your one-pager or your video? Why?

Written to an NFL general manager or a college recruiting director, your choice of team.

BEFORE YOU PUSH ON MONDAY, OCTOBER 26

Commit your skill, your one-pager or deck, and your script.

Answer every question in BRIEF.md, in your own words.

Run /audit, then /quiz-me.

Upload your repository's link to Canvas by Monday, October 26 at 11:59pm.